

$$\begin{aligned}
 \frac{1}{4} (\|u+v\|^2 - \|u-v\|^2) &= \frac{1}{4} (\langle u+v, u+v \rangle - \langle u-v, u-v \rangle) \\
 &= \frac{1}{4} (\langle u, u \rangle + \langle u, v \rangle + \langle v, u \rangle + \langle v, v \rangle - \langle u, u \rangle - \langle -v, u \rangle - \\
 &\quad - \langle u, -v \rangle - \langle -v, -v \rangle) = \frac{1}{4} (\langle u, v \rangle + \overline{\langle u, v \rangle} + \langle u, v \rangle \\
 &\quad + \langle u, v \rangle) = \frac{1}{4} (4 \cdot \text{Re}(\langle u, v \rangle)) = \underline{\text{Re}(\langle u, v \rangle)}
 \end{aligned}$$

však i jiné skokový nechtí, lineárníta druhé písmě
 a to že $\overline{-1} = -1$ a tedy $\langle -u, v \rangle = \langle u, v \rangle$

$$\begin{aligned}
 \frac{1}{4} (\|u - iv\|^2 - \|u + iv\|^2) &= \frac{1}{4} (\langle u - iv, u - iv \rangle - \langle u + iv, u + iv \rangle) = \\
 &= \frac{1}{4} (\langle u, u \rangle + \langle u, -iv \rangle + \langle -iv, u \rangle + \langle -iv, -iv \rangle - \langle u, u \rangle - \langle u, iv \rangle - \langle iv, u \rangle - \\
 &\quad - \langle iv, iv \rangle) = \frac{1}{4} (-i\langle u, v \rangle + i\langle v, u \rangle - i\langle u, v \rangle + i\langle v, u \rangle) = \\
 &= \frac{1}{2} i(\langle u, v \rangle + \overline{\langle u, v \rangle}) = \text{Im}(\langle u, v \rangle)
 \end{aligned}$$

pozor! jiné to same a vlastnost komplexního
 sdružení